In [1]:

```python
import pandas as pd
```

In [15]:

```python
df=pd.DataFrame({
    "Name":['Amit','Rahul','Amit','Sneha'],
    "Age":[25,30,25,28]
})
```

In [17]:

```python
df
```

Out[17]:

	Name	Age
0	Amit	25
1	Rahul	30
2	Amit	25
3	Sneha	28

In [19]:

```python
df.duplicated().sum()
```

Out[19]:

```
1
```

In [21]:

```python
#df.drop_duplicates()
```

In [23]:

```python
df.drop_duplicates(subset=['Name'])
```

Out[23]:

	Name	Age
0	Amit	25
1	Rahul	30
3	Sneha	28

In [25]:

```python
df=pd.DataFrame({
    "Name":['Amit','Rahul','Amit','Sneha'],
    "Age":[25,30,25,None]
})
```

In [27]:

```python
df
```

Out[27]:

	Name	Age
0	Amit	25.0
1	Rahul	30.0
2	Amit	25.0
3	Sneha	NaN

In [29]:

```python
df.isna().sum()
```

Out[29]:

```
Name    0
Age     1
dtype: int64
```

In [35]:

```python
x=df['Age'].mean()
```

In [39]:

```python
df['Age']=df['Age'].fillna(x)
```

In [41]:

```python
df
```

Out[41]:

	Name	Age
0	Amit	25.000000
1	Rahul	30.000000
2	Amit	25.000000
3	Sneha	26.666667

In [116…

```python
from math import nan
df=pd.DataFrame({
    "Name":['Amit',nan,'Amit','Sneha'],
    "Age":[25,30,25,None]
})
```

In [118…

```python
df
```

Out[118…

	Name	Age
0	Amit	25.0
1	NaN	30.0
2	Amit	25.0
3	Sneha	NaN

In [96]:

```python
df.isna().sum()
```

Out[96]:

```
Name    1
Age     0
dtype: int64
```

In [120…

```python
df
```

Out[120…

	Name	Age
0	Amit	25.0
1	NaN	30.0
2	Amit	25.0
3	Sneha	NaN

In [122…

```python
for i in df.columns:
    if df[i].dtypes=="object":
        x=df[i].mode()[0]
        df[i]=df[i].fillna(x)
    else:
        x=df[i].mean()
        df[i]=df[i].fillna(x)
```

In [124…

```python
df.isna().sum()
```

Out[124…

```
Name    0
Age     0
dtype: int64
```

In [126…

```python
df
```

Out[126…

	Name	Age
0	Amit	25.000000
1	Amit	30.000000
2	Amit	25.000000
3	Sneha	26.666667

Label Encoding

In [169…

```python
df=pd.DataFrame({
    "Name":['Amit','Rahul','Amit','Sneha'],
    "Gender":['Male','Male','Male','Female'],
    "City":["Pune","Mumbai","Pune","Delhi"]
})
```

In [171…

```python
df
```

Out[171…

	Name	Gender	City
0	Amit	Male	Pune
1	Rahul	Male	Mumbai
2	Amit	Male	Pune
3	Sneha	Female	Delhi

In [135…

```python
from sklearn.preprocessing import LabelEncoder

le=LabelEncoder()
```

In [137…

```python
df['Gender']=le.fit_transform(df['Gender'])
```

In [139…

```python
df
```

Out[139…

	Name	Gender	City
0	Amit	1	Pune
1	Rahul	1	Mumbai
2	Amit	1	Pune
3	Sneha	0	Delhi

In [143…

```python
le.classes_
```

Out[143…

```
array(['Female', 'Male'], dtype=object)
```

In [147…

```python
columns=df.columns
```

In [149…

```python
for i in df.columns:
    df[i]=le.fit_transform(df[i])
```

In [151…

```python
df
```

Out[151…

	Name	Gender	City
0	0	1	2
1	1	1	1
2	0	1	2
3	2	0	0

In [187…

```python
for i in df.columns:
    df[i]=le.fit_transform(df[i])
    print(i,le.classes_)
```

```
Name [0 1 2]
Gender [0 1]
City [0 1 2]
```

In [155…

```python
a=20
```

In [157…

```python
a
```

Out[157…

```
20
```

In [159…

```python
a=30
```

In [161…

```python
a
```

Out[161…

```
30
```

In [163…

```python
le.classes_
```

Out[163…

```
array([0, 1, 2])
```

In [173…

```python
df['Name']=le.fit_transform(df['Name'])
```

In [175…

```python
df
```

Out[175…

	Name	Gender	City
0	0	Male	Pune
1	1	Male	Mumbai
2	0	Male	Pune
3	2	Female	Delhi

In [177…

```python
le.classes_
```

Out[177…

```
array(['Amit', 'Rahul', 'Sneha'], dtype=object)
```

In [179…

```python
df['Gender']=le.fit_transform(df['Gender'])
```

In [181…

```python
df
```

Out[181…

	Name	Gender	City
0	0	1	Pune
1	1	1	Mumbai
2	0	1	Pune
3	2	0	Delhi

In [183…

```python
le.classes_
```

Out[183…

```
array(['Female', 'Male'], dtype=object)
```

In [ ]: